

ECE 315: Signals and Systems

Final Exam Review and Sample Questions

Instructor: Dongsheng Ding

May 4, 2026

The Final Exam will cover all lectures, homework, and midterm exams (and sample exams), with an emphasis on the material after Midterm II (about 60%). This guide is intended for self-study in preparation for the Final Exam. It summarizes the key concepts covered in class and includes self-check questions. A few sample questions on the material after Midterm II are provided. Solutions to the sample questions will not be released.

Below is a summary of key concepts and accompanying self-check questions.

- Basics of discrete- and continuous-time signals: discrete vs continuous, signal size, signal operation, periodicity, symmetry, examples: exponential, sinusoidal, impulse, step functions
- Basics of discrete- and continuous-time systems: interconnection, system properties: memoryless, invertibility, causality, stability, time-invariance, linearity
- Discrete-time and continuous-time linear time-invariant systems: convolution and its properties, impulse response, step response, system stability, block diagram
- Fourier series of continuous-time periodic signals: representation, convergence conditions, properties of Fourier series, Parseval's theorem, LTI system response to periodic signals
- Fourier series of discrete-time periodic signals: representation, properties of Fourier series, Parseval's theorem, LTI system response to periodic signals
- Fourier transform of continuous-time signals: representation, convergence conditions, properties of continuous-time Fourier transform, Parseval's theorem, LTI system response
- Fourier transform of discrete-time signals: representation, convergence conditions, properties of discrete-time Fourier transform, Parseval's theorem, LTI system response

- Bode plots of frequency response of continuous-time and discrete-time LTI systems
1. (**Ch. 1 of SS: Basic signals, signal operations**) Consider the standard signals: unit step, impulse, complex exponential, in discrete time and continuous time.
 - Define all these signals and sketch them;
 - Determine the relationships between unit step and impulse signals;
 - Define and sketch their transformed versions after time-shifting, time-scaling, and time-reversing.
 2. (**Ch. 1 of SS: Signal property: periodicity**) Consider the signals: $x[n]$ and $x(t)$, in discrete time and continuous time.
 - Define the periodicity conditions for them;
 - Define the average power and energy of them;
 - Assume they are periodic, define their average power and energy;
 - Determine the periodicity conditions for the sum, product, and division of two periodic signals.
 3. (**Ch. 1 of SS: Signal property: symmetry**) Consider the signals: $x[n]$ and $x(t)$, in discrete time and continuous time.
 - Determine the conditions when they are even and odd;
 - Find the even or odd components of them.
 4. (**Ch. 1 of SS: System properties**) Consider a discrete-time or continuous-time system with input and output $(x[n], y[n])$ or $(x(t), y(t))$.
 - Determine the conditions when they systems are memoryless, causal, linear, time-invariant, BIBO stable, and invertible;
 - Define linear time-invariant (LTI) systems.
 5. (**Ch. 2 of SS: LTI systems: impulse response**) Consider a discrete-time or continuous-time LTI system with input and output $(x[n], y[n])$ or $(x(t), y(t))$.
 - Define the impulse responses (i.e., $h[n]$ or $h(t)$) of LTI systems;
 - Determine the conditions on impulse responses when the LTI systems are memoryless, causal, BIBO stable, and invertible.
 6. (**Ch. 2 of SS: LTI systems: convolution**) Consider a discrete-time or continuous-time LTI system with input and output $(x[n], y[n])$ or $(x(t), y(t))$.

- Determine the discrete-time or continuous-time LTI system response for input $x[n]$ or $x(t)$;
 - Define the commutative, distributive, associative properties of convolution, and draw their block diagrams;
 - Define difference or differential equations as a description of discrete-time or continuous-time LTI systems.
7. (**Ch. 3 of SS:** Fourier series representations of periodic signals) Consider continuous-time and discrete-time periodic signals $x(t)$ and $x[n]$.
- Define their Fourier series coefficients and representations;
 - Understand harmonics, fundamental frequency, spectrum, magnitude and phase of spectrum;
 - Determine the conditions under which Fourier series representations ‘represent’ the original signals. Explain the mathematical meaning of ‘represent.’
 - Derive the properties of Fourier series representations: linearity, time-shifting, time-reversal, time-scaling, multiplication, differentiation, and Parseval’s theorem (Tables 3.1 and 3.2 of SS).
 - Compute the Fourier series coefficients for some basic signals: sinusoids, complex exponentials, periodic square waves, and periodic impulse trains (Tables 4.2 and 5.2 of SS).
8. (**Ch. 3 of SS:** LTI systems: periodic signal response) Consider a discrete-time or continuous-time LTI system with impulse responses $h[n]$ or $h(t)$.
- Define the LTI system functions or eigenvalues of the LTI systems;
 - Determine the LTI system responses to periodic input signals $x[n]$ or $x(t)$.
9. (**Ch. 4 of SS:** Fourier transforms of continuous-time aperiodic signals) Consider a continuous-time aperiodic signal $x(t)$.
- Define its Fourier transform pair;
 - Understand spectrum, magnitude and phase of spectrum;
 - Determine the conditions under which Fourier representation ‘represent’ the original signal. Explain the mathematical meaning of ‘represent.’
 - Derive the properties of continuous-time Fourier transform: linearity, time-/frequency-shifting, time-reversal, time-/frequency-scaling, conjugation, convolution, multiplication, differentiation, and Parseval’s theorem (Table 4.1 of SS).

- Compute the continuous-time Fourier transform for some basic signals: sinusoids, complex exponentials, periodic square waves, and periodic impulse trains (periodic signals); impulse, sinc, rectangular pulse, and decaying exponentials (aperiodic signals) (Table 4.2 of SS).
10. (**Ch. 4 of SS:** Continuous-time LTI systems: system response) Consider a continuous-time LTI system with impulse response $h(t)$.
- Determine the LTI system response to an aperiodic input $x(t)$ using Fourier transform;
 - Understand the frequency response and under what conditions on impulse response $h(t)$ it exists; understand low-pass, high-pass, band-pass filters;
 - Repeat the first task when the LTI systems are represented by differential equations.
11. (**Ch. 5 of SS:** Fourier transforms of discrete-time aperiodic signals) Consider a discrete-time aperiodic signal $x[n]$.
- Define its Fourier transform pair;
 - Understand spectrum, magnitude and phase of spectrum;
 - Determine the conditions under which Fourier representation ‘represent’ the original signal. Explain the mathematical meaning of ‘represent.’
 - Derive the properties of discrete-time Fourier transform: linearity, time-/frequency-shifting, time-reversal, time-/frequency-scaling, conjugation, convolution, multiplication, differentiation, and Parseval’s theorem (Table 5.1 of SS).
 - Compute the discrete-time Fourier transform for some basic signals: sinusoids, complex exponentials, periodic square waves, and periodic impulse trains (periodic signals); impulse, sinc, rectangular pulse, and decaying exponentials (aperiodic signals) (Table 5.2 of SS).
12. (**Ch. 5 of SS:** Discrete-time LTI systems: system response) Consider a discrete-time LTI system with impulse response $h[n]$.
- Determine the LTI system response to an aperiodic input $x[n]$ using Fourier transform;
 - Understand the frequency response and under what conditions on impulse response $h[n]$ it exists; understand low-pass, high-pass, band-pass filters;
 - Repeat the first task when the LTI systems are represented by difference equations.

13. (**Ch. 6 of SS: Bode plots**) Consider a discrete-time or continuous-time LTI system that is described by first-order and simple second-order differential or difference equations.

- Determine their frequency responses;
- Sketch their magnitude and phase in Bode plots.

Below is a set of sample questions on the material after Midterm II.

1. Consider the following continuous-time signals:

- $x(t) = \sin(t)$;
- $x(t) = \begin{cases} 1-t & 0 \leq t \leq 1 \\ t-1 & 1 \leq t \leq 2 \\ 0 & \text{otherwise.} \end{cases}$

Compute the Fourier transform $X(jw)$ for each continuous-time signal. Determine which type of convergence holds for the Fourier representation and justify your choice:

- ☐ Convergence everywhere
- ☐ Convergence with no energy difference
- ☐ Convergence except for discontinuities

2. Consider the following continuous-time causal and stable linear-time invariant system with input $x(t)$ and output $y(t)$ that is described by a differential equation:

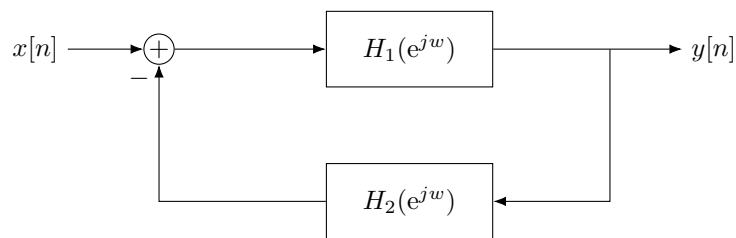
$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = x(t).$$

- (i) Determine the impulse response $h(t)$ of this system;
- (ii) Compute the output of the LTI system for the input $x(t) = e^{-t}u(t)$;
- (iii) Determine the frequency response $H(jw)$ of this system and sketch its Bode plot.
- (iv) Compute the output of the LTI system for the input $x(t) = \sin(t)$.

3. Consider a feedback system with two frequency responses

$$H_1(e^{jw}) = \frac{1}{1 + \frac{1}{2}e^{-jw}}$$

$$H_2(e^{jw}) = \frac{1}{16}e^{-j2w}$$



- (i) Determine the frequency response $H(e^{jw})$ of this feedback system and sketch its Bode plot;
 - (ii) Find the difference equation describing the input and output $x[n]$ and $y[n]$;
 - (iii) Determine the impulse response of the feedback system.
4. Consider an ideal discrete-time low-pass filter with frequency response,

$$H(e^{jw}) = \begin{cases} 1 & |w| \leq \frac{\pi}{3} \\ 0 & \frac{\pi}{3} < |w| \leq \pi \end{cases}$$

which extends to $\pm\infty$ periodically with period 2π .

- (i) Determine the impulse response $h[n]$ of this filter;
- (ii) Find the output if the input is given by

$$x[n] = 1 + 2 \cos\left(\frac{\pi}{4}n\right) + 3 \sin\left(\frac{5\pi}{3}n + \frac{\pi}{3}\right).$$